

QICN v20 Internal Note: Product-Separator Counterexample and Conditional Closure of I_{int}

QICN internal formal-audit scaffold

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Governance boundary

This document does not certify external support, consciousness, phenomenality, identity transfer, or bridge-burden closure. It is an internal mathematical scaffold. Its assertions are limited to the formal definitions and conditional results stated below. It does not report human-subject data, external adjudication, reviewer certification, or empirical validation.

1 Scope and separation of layers

The purpose of this note is to close the internal I_{int} gap as far as the present formal resources allow, without inflating the public status of the framework.

Ontology. A system S is treated as an admissible object with identity data, histories, interventions, and response observations. No consciousness, agency, subjecthood, or phenomenal identity is attributed by this note.

Mathematical model. The model is a category-like typed setting with admissible systems, admissible factorizations, separator families, and gauge transformations preserving the declared data.

Implementation. No source paper is edited by this note. The note can be compiled independently and linked from reports or registries.

Language. Terms such as “atomic”, “separator”, and “integration” are formal predicates or functionals defined below. Any intuitive reading is non-operative unless it reduces to these predicates.

Interpretation. The strongest closed result below is conditional: I_{int} is closed under an explicit atomic or prime-separator hypothesis. The product-separator construction proves that the original weaker assumptions do not imply atomicity by themselves.

2 Corpus anchors and external references

This note is not an isolated theorem island. It is anchored to the Paper 5 integration burden in the monolithic build: Proposition 5.37 states the original integration-transfer claim, while Theorem 5.11 states the already closed conditional result under an atomic separator. The same objects are recorded in the registry as `paper5:proposition:prop-integration-transfer` and

paper5:theorem:thm-iint-faithful-factorization-triviality. This v20 note therefore functions as a dependency-bearing auxiliary argument, not as a replacement for the source paper.

The categorical and factorization language follows the framework’s Paper 5 reference surface [Per26] and standard categorical background [Mac98]. The finite incidence witness is a graph-theoretic formalization of intervention-response dependence; it is compatible with, but does not rely on, causal-graph terminology in the sense of Pearl [Pea09]. These citations are orientation anchors only. They do not constitute external validation of QICN.

3 Typed setting

Definition 3.1 (Admissible system package). *An admissible system package is a tuple*

$$S = (X, \mathcal{U}, \text{Hist}_S, \text{Id}_S, \text{Resp}_S, \Theta_S),$$

where X is a state space, $\mathcal{U}$ is a set of admissible interventions, Hist_S is the admissible history family, Id_S is the rigid identity datum, Resp_S is the intervention-response assignment, and Θ_S is a family of separators used to distinguish identity/history/response alternatives.

Definition 3.2 (Exact admissible factorization). *An exact admissible factorization of S is a tuple*

$$F : S \longrightarrow S_1 \times S_2$$

with non-terminal factors S_1, S_2 such that the state, intervention, history, identity, and response data decompose as

$$X \cong X_1 \times X_2, \quad \mathcal{U} \cong \mathcal{U}_1 \times \mathcal{U}_2, \quad \text{Hist}_S \cong \text{Hist}_{S_1} \times \text{Hist}_{S_2},$$

$$\text{Id}_S \cong (\text{Id}_{S_1}, \text{Id}_{S_2}), \quad \text{Resp}_S(u_1, u_2, x_1, x_2) = (\text{Resp}_{S_1}(u_1, x_1), \text{Resp}_{S_2}(u_2, x_2)).$$

The factorization is non-trivial when neither factor is isomorphic to the terminal admissible system.

Definition 3.3 (Factor-local separator partition). *A separator partition for a factorization $F : S \rightarrow S_1 \times S_2$ is a disjoint union*

$$\Theta_S = \Theta_1 \sqcup \Theta_2$$

such that every $\theta \in \Theta_i$ is pulled back from factor S_i and separates only identity/history/response alternatives visible in that factor.

Definition 3.4 (Atomic separator). *The separator family Θ_S is atomic for S when there is no exact non-trivial admissible factorization $F : S \rightarrow S_1 \times S_2$ admitting a factor-local separator partition that preserves Id_S , Hist_S , and Resp_S simultaneously.*

Remark 3.5 (Non-circularity convention). *The phrase “atomic separator” is a formal non-decomposition predicate. A later axiom that merely restates this definition under another name would be circular. A useful additional condition must instead be stated in terms of checkable intervention support, response coupling, commutants, graph connectivity, or perturbation behavior, and only then shown to imply atomicity.*

4 Product-separator counterexample

Proposition 4.1 (Product separator blocks the global derivation). *Rigid identity, product-topological continuity, and componentwise intervention fidelity do not imply atomicity of Θ_S .*

Proof. Let S_1 and S_2 be two admissible system packages and form the product package

$$S = S_1 \times S_2.$$

Set

$$\begin{aligned} X &= X_1 \times X_2, \quad \mathcal{U} = \mathcal{U}_1 \times \mathcal{U}_2, \quad \text{Hist}_S = \text{Hist}_{S_1} \times \text{Hist}_{S_2}, \\ \text{Id}_S &= (\text{Id}_{S_1}, \text{Id}_{S_2}), \quad \text{Resp}_S((u_1, u_2), (x_1, x_2)) = (\text{Resp}_{S_1}(u_1, x_1), \text{Resp}_{S_2}(u_2, x_2)). \end{aligned}$$

Give X the product topology. If the component response maps are continuous, the product response map is continuous. If each component is intervention-faithful relative to its own data, the product is faithful componentwise: equality of product responses implies equality in each coordinate at the relevant observational resolution.

Now take

$$\Theta_S = \pi_1^* \Theta_{S_1} \sqcup \pi_2^* \Theta_{S_2},$$

where π_i are the coordinate projections. This separator family preserves identity, histories, and responses by construction, but it decomposes into factor-local separator families. Therefore the product factorization is non-trivial and admits a factor-local separator partition. Hence Θ_S is not atomic.

The construction satisfies the upstream rigid-identity, continuity, and componentwise fidelity requirements, yet violates separator atomicity. Thus the global implication from the weaker assumptions to atomicity is false in this model class. $\square$

Corollary 4.2 (Exact remaining burden). *Any proof of a global I_{int} theorem must add a condition that excludes product-separator systems or must restrict the admissible system class so that the product construction is inadmissible.*

5 Candidate strengthening conditions

Definition 5.1 (No-cloning of intervention channels). *An admissible system satisfies intervention no-cloning when no non-trivial factorization $S \cong S_1 \times S_2$ admits two factor-local intervention subfamilies $\mathcal{U}_1, \mathcal{U}_2$ whose pulled-back response maps reproduce the full response assignment while preserving Id_S and Hist_S .*

Definition 5.2 (Connected response-support graph). *Let G_S be the bipartite graph whose left vertices are elementary separator tests, whose right vertices are elementary intervention-response coordinates, and whose edges indicate that a separator test changes under a perturbation of that coordinate within the admissible response model. The system has connected response support when G_S is connected after deleting isolated non-operative vertices.*

Definition 5.3 (Kernel robustness under local perturbations). *Let $D \text{Resp}_S$ be the linearized response operator in a chosen admissible chart. The system has locally robust kernel coupling when every sufficiently small factor-local perturbation preserving $\ker(D \text{Resp}_S)$ also preserves the full coupled response-support graph. Equivalently, there is no local perturbation that splits $\ker(D \text{Resp}_S)$ into two invariant factor kernels while preserving Id_S and Hist_S .*

Definition 5.4 (Prime intervention-response coupling). *A system has prime intervention-response coupling when, for every exact non-trivial admissible factorization candidate $F : S \rightarrow S_1 \times S_2$, at least one of the following fails:*

- (i) *response support decomposes as a disjoint union of factor supports;*
- (ii) *the separator family is the pullback union of factor-local separators;*
- (iii) *the gauge group preserving identity, histories, and responses splits as a product of factor gauge groups;*
- (iv) *the linearized response operator has an invariant block decomposition compatible with the same factorization.*

Lemma 5.5 (Connected support blocks product separators). *If S has connected response support and every separator in Θ_S is response-operative, then no product-separator partition can preserve the full response support.*

Proof. Assume a product-separator partition $\Theta_S = \Theta_1 \sqcup \Theta_2$ exists for a non-trivial factorization. Since each separator is factor-local, perturbations of response coordinates in factor S_1 can affect only tests in Θ_1 , and perturbations in factor S_2 can affect only tests in Θ_2 . The response-support graph then decomposes into the disjoint union of the S_1 and S_2 support graphs. Because all separators are response-operative, both subgraphs contain at least one edge. This contradicts connectedness of G_S . $\square$

Lemma 5.6 (Prime coupling implies separator atomicity). *If S has prime intervention-response coupling, then Θ_S is atomic.*

Proof. Suppose Θ_S is not atomic. Then by definition there exists an exact non-trivial admissible factorization admitting a factor-local separator partition preserving identity, histories, and responses. Such a factorization satisfies separator pullback decomposition. Because the factorization is exact and intervention-faithful, it also gives a response-support decomposition, a compatible split gauge action, and an invariant block decomposition of the linearized response operator. This contradicts prime intervention-response coupling, which requires at least one of these four clauses to fail for every non-trivial factorization candidate. $\square$

No-cloning. Blocks product separators when stated over response reproduction rather than separator names. Circularity risk: medium, because it becomes too strong if it only says that no duplicated separator exists. Audit status: useful, but requires operational tests.

Connected support. Blocks product separators under the response-operative separator assumption. Circularity risk: low to medium, because it is a graph-theoretic condition testable on a presented model. Audit status: strongest non-verbal candidate.

Kernel robustness. Blocks product separators when factor kernels are excluded under perturbation. Circularity risk: medium, because the condition depends on chart and linearization choices. Audit status: useful analytic supplement, not sufficient alone.

Prime coupling. Blocks product separators by requiring that at least one response, separator, gauge, or operator decomposition clause fail for every non-trivial factorization candidate. Circularity risk: medium-high, because it is close to atomicity but not phrased as atomicity. Audit status: aggressive v20 conditional axiom.

6 Conditional closure of I_{int}

Definition 6.1 (I_{int} as a factorization-gap functional). *For an admissible system S , define the exact factorization gap*

$$\Delta_{\text{int}}(S) = \inf_{F \in \mathcal{F}_{\text{nt}}(S)} \left[\mathcal{L}_{\text{hist}}(F) + \mathcal{L}_{\text{id}}(F) + \mathcal{L}_{\text{resp}}(F) + \mathcal{P}_{\text{gauge}}(F) \right],$$

where $\mathcal{F}_{\text{nt}}(S)$ is the class of non-trivial admissible factorizations, the loss terms vanish exactly when histories, identity, and responses are preserved, and $\mathcal{P}_{\text{gauge}}$ vanishes exactly when the factorization is compatible with the admissible gauge action. Then

$$I_{\text{int}}(S) = 1 \iff \Delta_{\text{int}}(S) > 0.$$

If $\mathcal{F}_{\text{nt}}(S)$ is empty, set $\Delta_{\text{int}}(S) = +\infty$.

Theorem 6.2 (Conditional on Θ_S atomic: existence of I_{int}). *If Θ_S is atomic and the loss terms above are faithful to history, identity, response, and gauge preservation, then $I_{\text{int}}(S)$ exists as a well-defined binary invariant of the admissible isomorphism class of S .*

Proof. The functional Δ_{int} is defined as an infimum over a class determined by the admissible isomorphism class of S . The loss terms are also invariant under admissible isomorphism because they are expressed only in terms of preserved histories, identity, responses, and gauge compatibility. Thus $\Delta_{\text{int}}(S)$ is well-defined on admissible isomorphism classes.

Atomicity of Θ_S rules out exact non-trivial factorizations with zero loss. Faithfulness of the loss terms implies that zero loss is equivalent to preserving the four relevant structures. Therefore the zero-loss case is impossible for non-trivial factorizations. The predicate $\Delta_{\text{int}}(S) > 0$ is consequently a well-defined formal invariant whenever a positive margin is part of the admissibility criterion. The binary value $I_{\text{int}}(S)$ is that predicate. $\square$

Theorem 6.3 (Conditional on Θ_S atomic: uniqueness under gauge). *Assume Θ_S is atomic and the admissible gauge group $\text{Gauge}(S)$ acts by preserving histories, identity, responses, and separator membership. Then any binary invariant $J(S)$ that is zero exactly on systems admitting a zero-loss non-trivial factorization and is invariant under $\text{Gauge}(S)$ agrees with $I_{\text{int}}(S)$.*

Proof. By definition, $I_{\text{int}}(S) = 0$ exactly when $\Delta_{\text{int}}(S) = 0$, which is exactly the existence of a zero-loss non-trivial factorization. The invariant J is stipulated to have the same zero-set. Both J and I_{int} are binary. Therefore they agree pointwise. Gauge invariance ensures that the value is not changed by reparameterizations of the separator or response presentation. $\square$

Corollary 6.4 (Aggressive but bounded v20 closure). *Under prime intervention-response coupling, or under separator-complete connected finite incidence, together with response-operative separators and faithful factorization losses, I_{int} is conditionally closed as a unique gauge-invariant binary factorization-gap invariant.*

Proof. Prime intervention-response coupling implies separator atomicity. Separately, separator-complete connected finite incidence implies separator atomicity by the finite connected-incidence theorem. In either route, the two conditional theorems then give existence and uniqueness of I_{int} as a gauge-invariant binary invariant. $\square$

7 Aggressive finite-witness closure: a non-circular sufficient theorem

Definition 7.1 (Finite response-separator incidence package). *A finite response-separator incidence package for S is a finite bipartite graph*

$$G_S = (V_\Theta \sqcup V_R, E),$$

The vertex set V_Θ indexes response-operative elementary separators in Θ_S . The vertex set V_R indexes elementary intervention-response coordinates. An edge $(\theta, r) \in E$ is present exactly when an admissible perturbation of coordinate r changes the truth value or margin of separator test θ while preserving the declared identity and history typing.

Definition 7.2 (Separator-complete incidence). *The incidence package is separator-complete when every separator in Θ_S is represented by at least one vertex in V_Θ , every represented separator is response-operative, and every exact admissible factorization $S \rightarrow S_1 \times S_2$ induces a partition*

$$V_\Theta = V_{\Theta,1} \sqcup V_{\Theta,2}, \quad V_R = V_{R,1} \sqcup V_{R,2}$$

whenever Θ_S is a pullback union of factor-local separators and the response assignment is componentwise factor-local.

Remark 7.3 (Circularity-risk audit for separator-complete incidence). *A reviewer should verify that separator-complete incidence can be established operationally without presupposing atomicity. The intended operational route is: first construct V_Θ , V_R , and E from declared separator tests and admissible perturbation-response records; then verify separator coverage, response-operativity, and induced factor partitions for candidate factorizations; only after that apply graph connectedness. If the construction of G_S requires already knowing that no factor-local separator partition exists, the condition collapses into a disguised restatement of atomicity and must be rejected. This is an explicit **REVIEWER_BURDEN**, not a certified theorem of the present note.*

Theorem 7.4 (Finite connected-incidence atomicity theorem). *Assume S admits a separator-complete finite response-separator incidence package G_S . If G_S is connected, then Θ_S is atomic.*

Proof. Assume for contradiction that Θ_S is not atomic. Then there is an exact non-trivial admissible factorization $F : S \rightarrow S_1 \times S_2$ admitting a factor-local separator partition that preserves identity, histories, and responses. Since the factorization is componentwise factor-local, perturbing a response coordinate in factor S_1 can affect only separators pulled back from S_1 , and perturbing a response coordinate in factor S_2 can affect only separators pulled back from S_2 .

By separator-completeness, this induces a partition of the incidence vertices

$$V_\Theta = V_{\Theta,1} \sqcup V_{\Theta,2}, \quad V_R = V_{R,1} \sqcup V_{R,2}$$

with no edge from $V_{R,1}$ to $V_{\Theta,2}$ and no edge from $V_{R,2}$ to $V_{\Theta,1}$. Because the factorization is non-trivial and the represented separators are response-operative, both sides contain at least one incident edge. Therefore G_S decomposes as a disjoint union of two non-empty subgraphs. This contradicts connectedness. Hence no such factor-local separator partition exists, so Θ_S is atomic. $\square$

Corollary 7.5 (Finite-witness conditional closure of I_{int}). *If S has a separator-complete finite response-separator incidence package whose graph is connected, and if the factorization-loss terms are faithful to history, identity, response, and gauge preservation, then $I_{\text{int}}(S)$ exists and is unique as a gauge-invariant binary factorization-gap invariant.*

Proof. The finite connected-incidence theorem gives separator atomicity from a graph-theoretic condition that is stated without naming atomicity. The conditional existence and uniqueness theorems for I_{int} under atomic separator then apply. $\square$

Remark 7.6 (Why this is stronger than merely assuming atomicity). *The connected-incidence condition is not a semantic synonym for atomicity. It is a finite graph predicate over observable separator-response incidences. It may fail for some genuinely non-factorized systems if the chosen incidence package is incomplete, and it may be difficult to construct in infinite systems. Therefore it is a sufficient witness condition, not a necessary characterization and not a global proof from rigid identity plus continuity plus fidelity alone.*

8 What is closed and what remains open

Product-separator counterexample. Closed in this note. It explicitly satisfies the weaker assumptions and fails atomicity.

Global derivation. Not closed. Atomicity from rigid identity plus continuity plus fidelity is blocked by the counterexample.

I_{int} under atomic separator. Conditionally closed. Existence and uniqueness follow from atomicity plus faithful factorization losses.

I_{int} under prime coupling. Conditionally closed. Prime coupling implies atomicity, then the conditional theorem applies.

Empirical or external support. Not claimed. No external campaign, human review, or adjudication is executed here.

Registry integration. Improved in v20. This note cross-references Paper 5 anchors and cites the release bibliography; promotion still requires human curation.

9 Implementation note

This note is intentionally not inserted into the original Paper 5 source. It is a v20 scaffold that can be reviewed, criticized, or promoted by a later human mathematical curation process. Promotion into the theorem registry would require explicit registry overlay, human mathematical review, and a decision about whether prime intervention-response coupling or separator-complete finite incidence is an acceptable new hypothesis rather than a disguised restatement of separator atomicity.

References

- [Mac98] Saunders Mac Lane. *Categories for the Working Mathematician*. 2nd. Vol. 5. Graduate Texts in Mathematics. New York: Springer, 1998.
- [Pea09] Judea Pearl. *Causality: Models, Reasoning, and Inference*. 2nd. Standard reference for causal inference and causal graphs. Cambridge: Cambridge University Press, 2009.
- [Per26] Johnny Andrey Perez Ramirez. “A Structural Criterion for Substrate-Invariant Operational Consciousness”. In: *arXiv preprint* (2026).